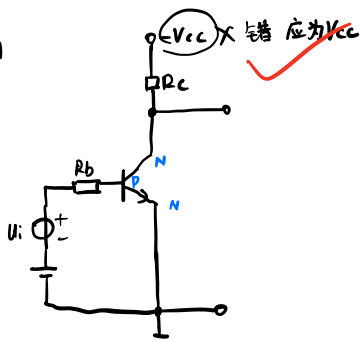
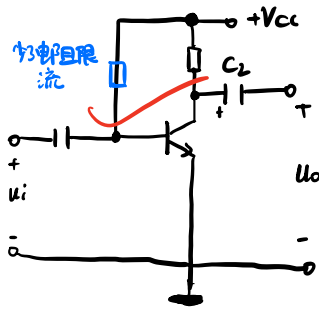


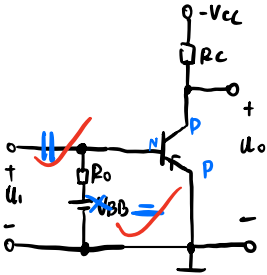
2.1
(a)



(b)



(c)



错一: 发射结反偏了

错二: 交流直接并入了
并未偏置

错误①

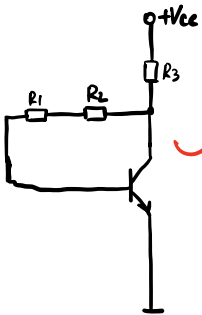
(d)

V_{BB} 和 V_{CC} 都应该加两个限流电阻
电容极性左边为 '+' / 为什么要加限流 (b.c. 为什么要加 e 极为什么就不加)

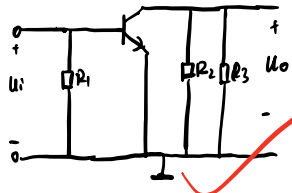
2.2

(a)

直流通路

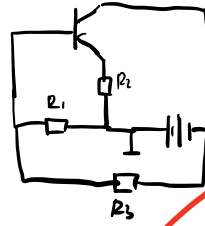


交流通路

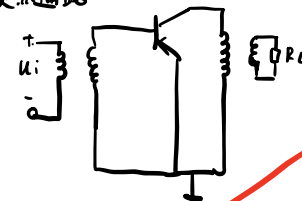


(d)

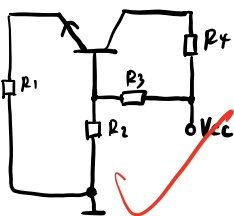
直流通路



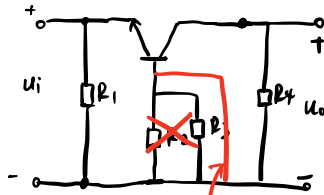
交流通路



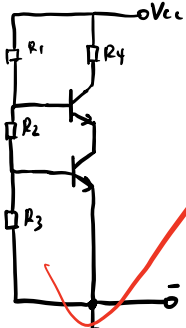
(b) 直流通路



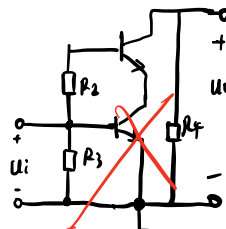
交流通路



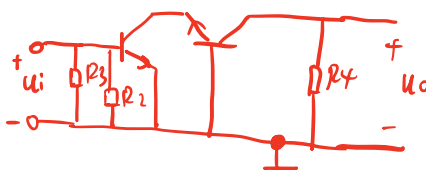
(c) 直流通路



交流通路



取最短路径接地
有导线直达地 绝不经电阻

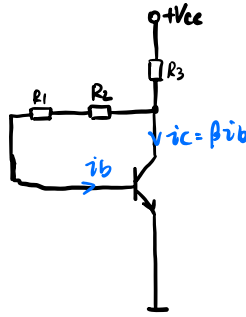


(a) 共射放大电路

(b) 共基放大电路

下面求取 Q 、 A_u 、 R_i 与 R_o

(a) 直流通路



$$V_{CC} = U_{BEQ} + i_b(R_1 + R_2) + (1 + \beta)i_b \cdot R_3$$

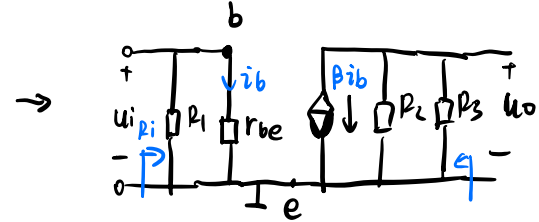
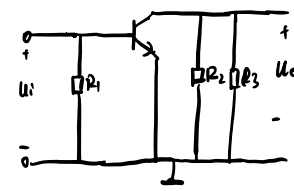
$$I_{BQ} = \frac{V_{CC} - U_{BEQ}}{R_1 + R_2 + (1 + \beta)R_3}$$

$$I_{EQ} = (1 + \beta)I_{BQ} = (1 + \beta) \frac{V_{CC} - U_{BEQ}}{R_1 + R_2 + (1 + \beta)R_3}$$

$$U_{CEQ} = V_{CC} - (1 + \beta)I_{BQ} \cdot R_3$$

$$R_{be} = R_{bb'} + (1 + \beta) \frac{U_T}{I_{EQ}}$$

交流通路



$$A_u = \frac{u_o}{u_i} = - \frac{\beta i_b \cdot R_2 // R_3}{i_b \cdot r_{be}} = - \beta \cdot \frac{R_2 // R_3}{r_{be}} \quad (\text{输出会反相})$$

$$R_i = R_1 // r_{be}$$

考虑 R_o

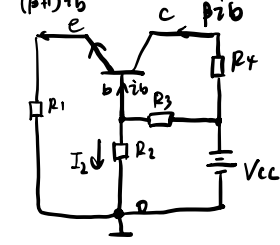
$$u_i = 0$$

$$i_b = 0$$

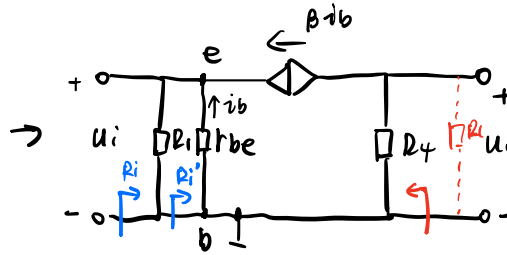
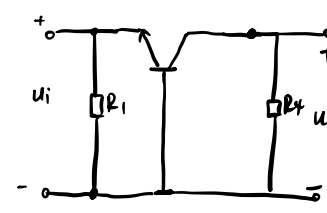
$\beta i_b = 0 \Rightarrow$ 断路

$$R_o = R_2 // R_3$$

(b) 直流通路



交流通路

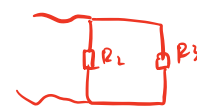


对基极那个点进行戴维南等效

$$V(b) = V_{CC} \times \frac{R_2}{R_2 + R_3} \Rightarrow \text{等效电压}$$

R_T : 电压源置0

$$R_T = R_2 // R_3$$

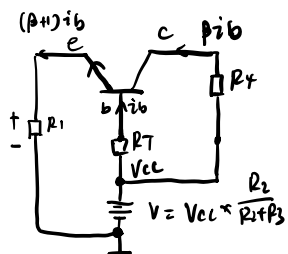


$$A_u = \frac{u_o}{u_i} = \frac{\beta i_b \cdot R_4}{i_b \cdot r_{be}} = \frac{\beta R_4}{r_{be}}$$

$$R_i' = \frac{1}{1 + \beta} \cdot r_{be}$$

$$R_i = R_1 // R_i'$$

$$R_o = R_4$$



$$0 + (\beta + 1) i_b \cdot R_1 + U_{BEQ} + i_b \cdot R_2 = V_{CC}$$

$$i_b [(\beta + 1) R_1 + R_2] = V_{CC} - U_{BEQ}$$

$$i_b = \frac{V_{CC} - U_{BEQ}}{(\beta + 1) R_1 + R_2 / \beta}$$

$$i_e = (\beta + 1) i_b = I_{EQ}$$

$$r_{be} = (\beta + 1) \cdot \frac{U_T}{I_{EQ}}$$

3

求 Ri 与 Ro 总结

2.4

$$1V > 0.7V$$

$$i_{EQ} = \frac{0.3V}{15k\Omega} = \frac{300mV}{15k\Omega} = 20\mu A$$

$$U_o' = V_{CC} - i_c \cdot R_C$$

$$U_o = \frac{R_L}{R_o + R_L} \times U_o'$$

$$U_o' = \frac{R_o + R_L}{R_L} U_o$$

$$i_c R_C = V_{CC} - \frac{R_o + R_L}{R_L} U_o$$

$$i_c = \frac{V_{CC}}{R_C} - \frac{U_o}{R_C} \cdot \frac{R_o + R_L}{R_L}$$

$$= 4mA - \frac{1}{3k\Omega} \cdot \frac{R_o + R_L}{R_L} U_o$$

$$U_o = U_{CE}$$

$$\text{when } R_L = \infty$$

$$i_c = 4mA - \frac{U_o}{3k\Omega}$$

$$\text{when } R_L = 3k\Omega$$

$$i_c = 4mA - \frac{U_o}{1.5k\Omega}$$

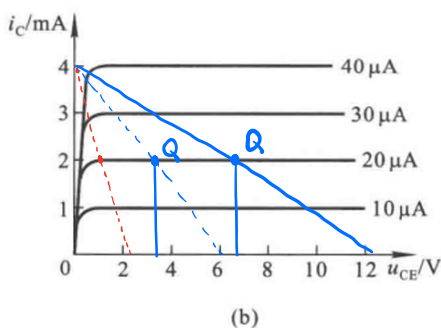
最大不失真输出电压 U_{om}

4

$$U_{omax} = U_{CEQ} - U_{CES} \approx U_{CEQ} - U_{BEQ}$$

$$= 2.3V$$

$$U_{om} = \frac{2.3V}{\sqrt{2}} = 1.63V$$

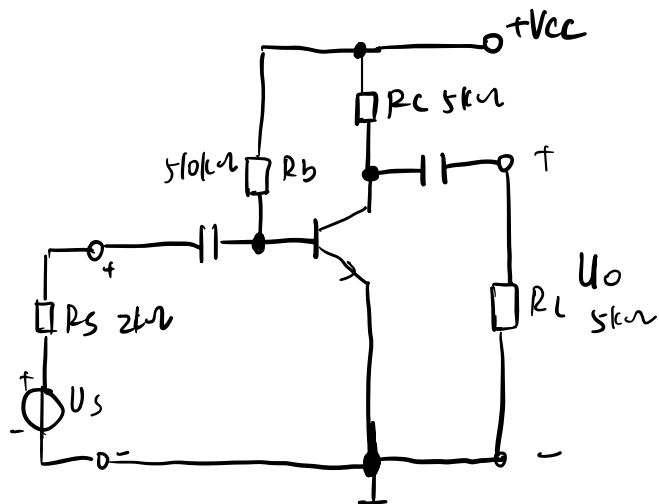


5 截止失真 vs 饱和失真
两者到底是什么?
区别到底是什么?

2.5

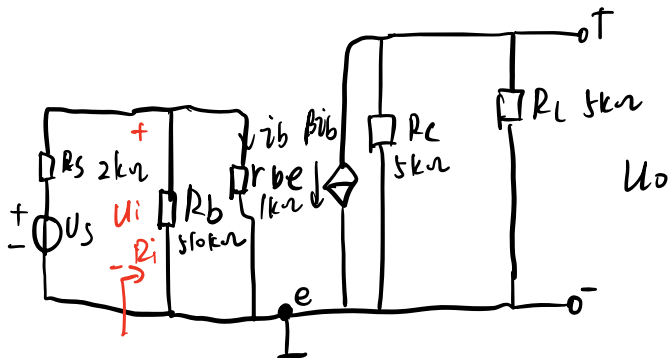
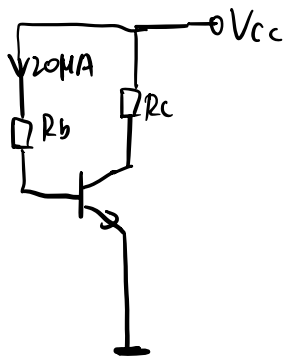
$\beta = 80$ 静态时 $U_{BEQ} = 0.7V$
 $r_{be} = 1k\Omega$ $U_{CEQ} = 4V$
 $U_i = 20mV$ $I_{BQ} = 20\mu A$

直流通路



直流通路

交流通路



$$\dot{A}_u = \frac{U_o}{U_i} = \frac{-2.5k\Omega \times \beta i_b}{1k\Omega \times i_b}$$

$$= -2.5 \times 80 = -200$$

5

为什么??? 是因为 A_u 这些是动态量不能用

静态电流 I_{BQ} 吗? $R_o = 5k\Omega$

(1) ~~✓~~ (4) ~~✓~~ (5) ~~✓~~

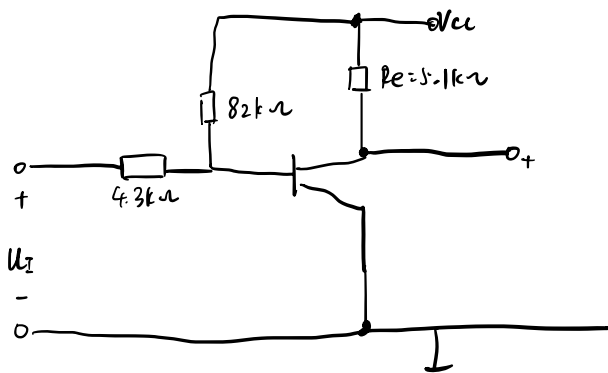
(8) ~~✓~~ (9) ~~✓~~ (12) ~~✓~~

(2) ~~✓~~ (3) ~~✓~~

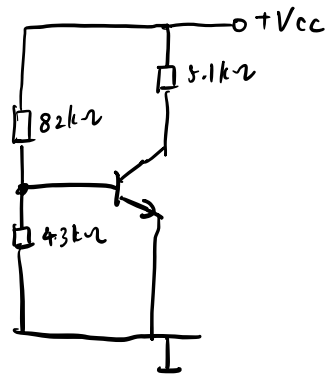
(6) ~~✓~~ (7) ~~✓~~ (10) ~~✓~~ (11) ~~✓~~

$$4\mu A \cdot 2k\Omega = 8mV$$

2.6



直流通路



$$i_b = \frac{15V - 0.7V}{82k\Omega} - \frac{0.7V}{4.3k\Omega} = 0.011599mA$$

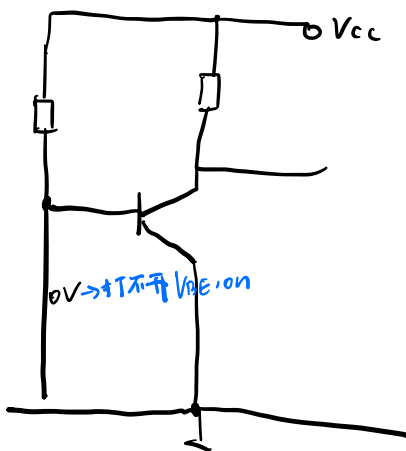
$$i_c = \beta i_b = 120 \times 0.011599mA = 1.391945547mA$$

$$V_c = 15V - 5k\Omega \times 1.391945547mA = 8.04027V$$

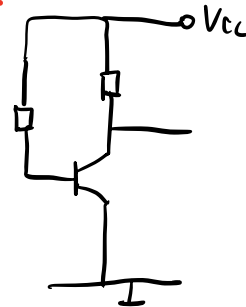
(2) R_{b1} 短路

三极管处于截止区

(3) R_{b1} 断路



$$V_c = V_{CC} = 15V$$



$$i_b = \frac{15V}{82k\Omega} = 0.1829mA$$

$$i_c = \beta i_b = 21.95219mA$$

$$V_c = 15V - 5.1k\Omega \times 21.95219mA$$

< 0

不成立

$\therefore$ 处于饱和区

$$V_c = 0.5V$$

(4) 截止状态

$$V_c = V_{CC} = 15V$$

(5) 三极管烧

$$V_c = 0.5V$$

难以判断

(6) $V_c = 15V$

错点: 饱和

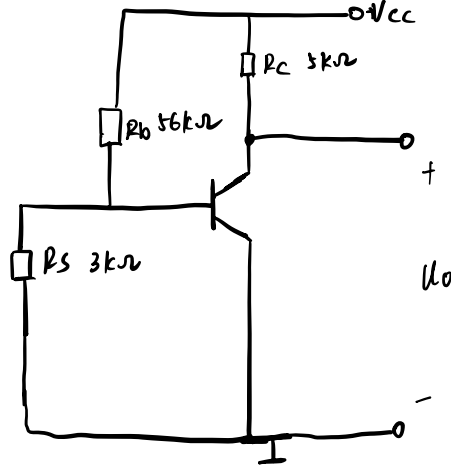
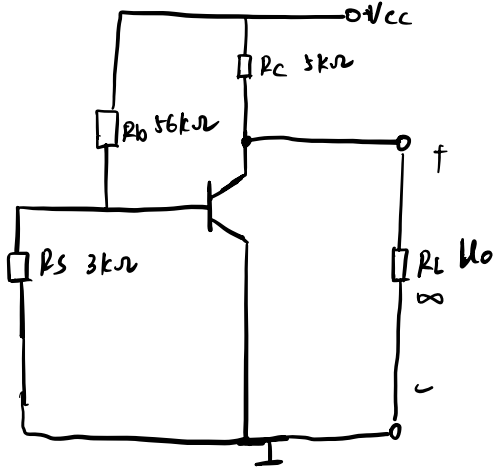
时集电极电位

$$= U_{CES}$$

而不是 $V_{CC} - U_{CES}$

2.7 $\beta = 80$ $r_{bb'} = 100$ $R_L = \infty$ $R_L = 3k\Omega$

直流通路

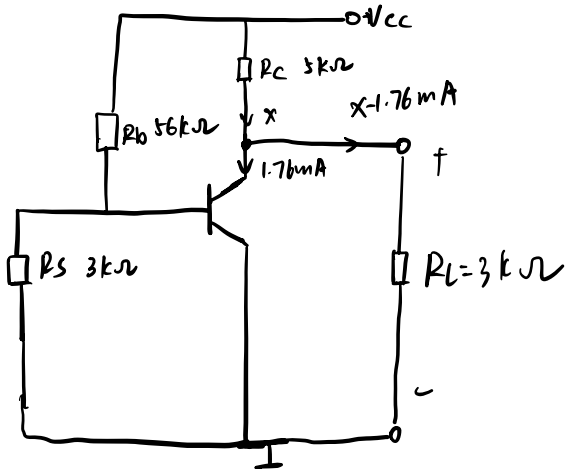


$R_L = \infty$ 时: $i_b = \frac{14.3V}{56k\Omega} - \frac{0.7V}{3k\Omega} = 22\mu A = I_{BQ}$

$i_c = \beta i_b = 1760\mu A = 1.76mA$

$U_{CEQ} = 15V - 5k\Omega \cdot 1.76mA = 6.2V$

直流通路



$i_b = 22\mu A$

$i_c = 1.76mA$

$5X + 3 \cdot (X - 1.76) = 15$

$8X = 15 + 3 \times 1.76$

$= 15 + 5.28$

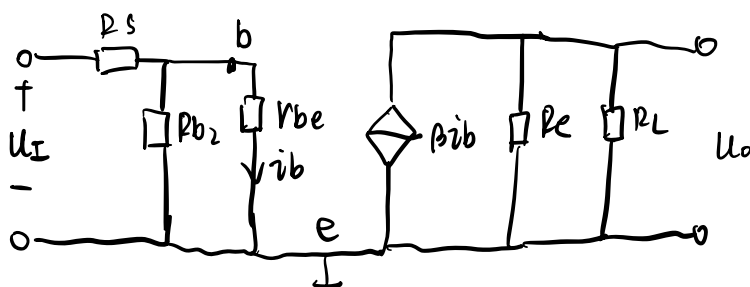
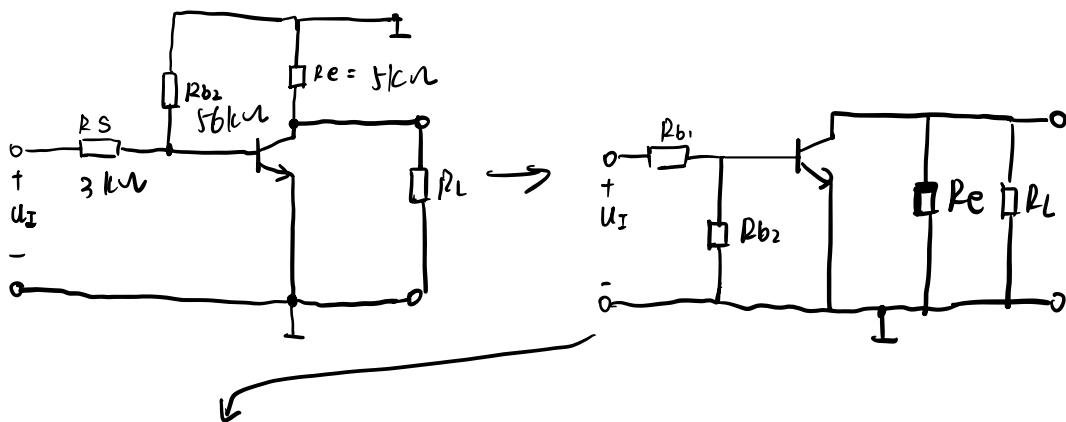
$= 20.28$

$X = 2.535mA$

$U_{CE} = 15V - 2.535mA \times 5k\Omega$

$= 2.325V$

交流通路



$$A_u = \frac{\beta i_b \cdot R_E // R_L}{i_b \cdot r_{be}}$$

$$r_{be} = r_{bb'} + (1 + \beta) \frac{U_T}{I_{EQ}}$$

当 $R_L = \infty$ 时

$$r_{be} = 100 + 81 \times \frac{26\text{mV}}{1.782\text{mA}} = 1281.8\Omega$$

$$A_u = - \frac{80 \times 5k\Omega}{1281.8\Omega} = -312$$

$$R_i = 1281.8\Omega$$

$$R_o = 5k\Omega$$

当 $R_L = 3k\Omega$ 时

$$r_{be} = 1281.8\Omega$$

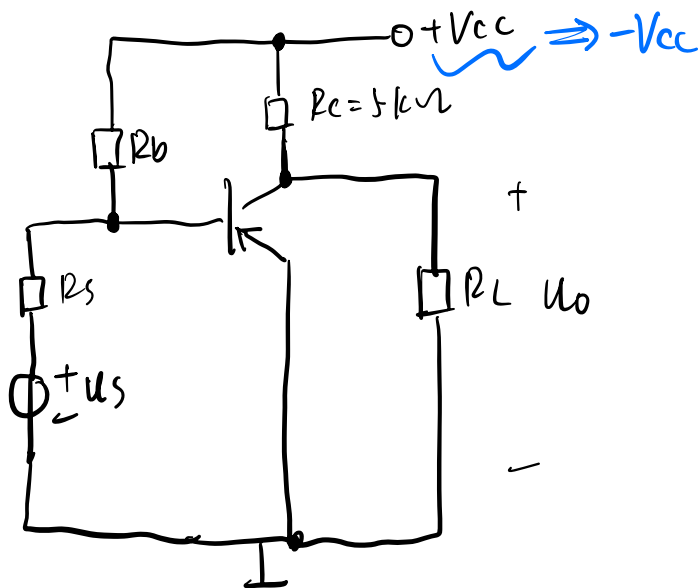
$$A_u = \frac{80 \times 3k\Omega \times \frac{5 \times 3}{5+3}}{1281.8\Omega}$$

$$= -117$$

$$R_i = 1281.8\Omega$$

$$R_o = 5k\Omega$$

2.8



Q点 U_{BEQ} , U_{CEQ} 数值不变但
极性翻转

A_u , R_i , R_o 不变

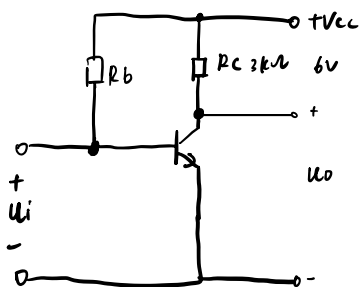
发生了截止失真

解决方法: 适当减小 R_b / 适当增大 V_{CC}
使Q点右移

2.9 (1)

$\beta = 100$ $r_{be} = 1k\Omega$

直流通路



$$i_C = \frac{6V}{3k\Omega} = 2mA$$

$$i_B = \frac{2mA}{100} = 20\mu A$$

$$\frac{12V}{i_B} = R_b = \frac{12V}{2 \times 10^{-2} mA} = 600k\Omega$$

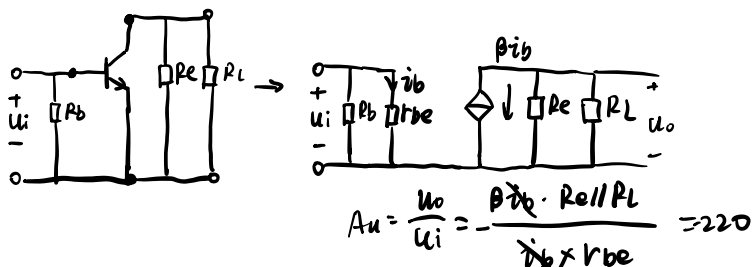
$$\frac{12V - 0.7V}{2 \times 10^{-2} mA} = 565k\Omega \quad \text{别忘了减去 } U_{BE(on)}$$

$$\therefore R_b \text{ 约为 } 565k\Omega$$

(2)

交流通路:

题目要求 $A_u > 220$



$$\frac{R_C \parallel R_L}{r_{be}} = 2.2 \quad \text{答案里数据对应过来是 } 2$$

$$R_C \parallel R_L = 2.2k\Omega$$

$$\frac{R_C \times 5}{R_C + 5} = 2.2$$

$$5R_C = 2.2R_C + 11$$

$$2.8R_C = 11 \quad R_C = \frac{11}{2.8} = \frac{55}{14}k\Omega$$

$$R_{E1} = 2k\Omega + 1k\Omega$$

$$R_E = 3.3k\Omega$$

2.10 不会

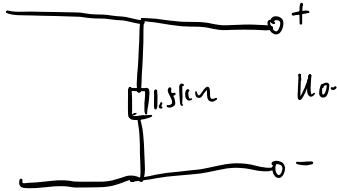
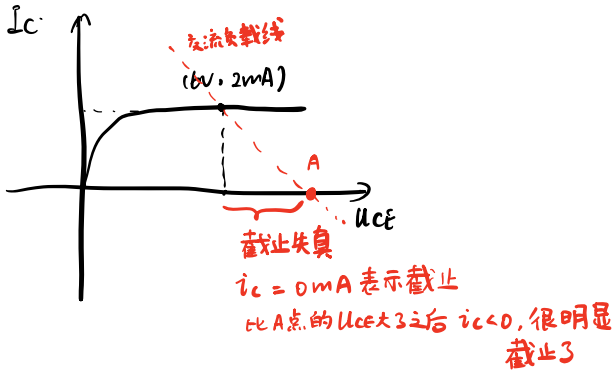
问：最大不失真输出电压要考虑哪些方面？

静态： $I_{CQ} = 2mA$

$$U_{CEQ} = 6V$$

$$I_{BQ} = 20\mu A$$

$$R_b = 565k\Omega$$



$$U_{om} = \frac{5.4V}{\sqrt{2}}$$

解：已知

$$U_{CEQ} = 6V$$

那么要分两种失真情况讨论

① 截止失真

$i_C = 0$ 时候的 U_O - 静态时的 U_O

$$2mA \times 1.5k\Omega = 3V$$

② 饱和失真

$$U_{om} = \frac{6V - 0.6V}{\sqrt{2}}$$

$\therefore$ 最大不失真输出电压为 $U_{om} = \frac{3}{\sqrt{2}} V$

想要最大化不失真输出电压

$$I_{CQ} \cdot 1.5k\Omega = U_{CEQ} - 0.6V$$

$$= 12V - I_{CQ} \cdot 3k\Omega - 0.6V$$

$$4.5k\Omega \cdot I_{CQ} = 11.4V$$

$$I_{CQ} = \frac{11.4}{4.5} mA = 2.53 mA$$

$$= \frac{38}{15} mA$$

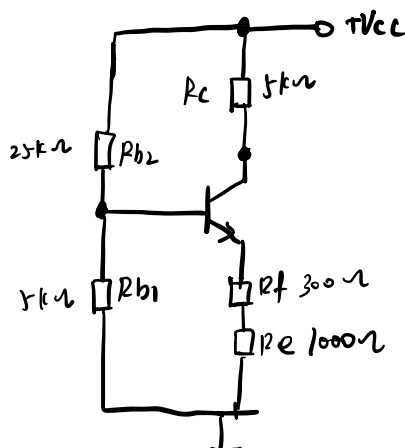
$$I_{BQ} = \frac{38}{15} \div 100 = \frac{38}{1500} = \frac{19}{750} mA$$

$$R_b = \frac{12V - 0.7V}{\frac{19}{750} mA} = 446.05k\Omega$$

$$U_{om} \approx \frac{3.8V}{\sqrt{2}} = 2.687V$$

2.11

(1) 直流通路



$$i_b = \frac{12V - 0.7V}{25k\Omega} - \frac{0.7V}{5k\Omega} = 0.312mA$$

错点：只有在发射极^{极支路}上没有 R_f, R_e 时候可以直接用 $0.7V$

$$I_{BQ} = 0.312mA$$

$$i_c = 31.2mA$$

一个技巧：若 $R_{b1} \parallel R_{b2} \ll (1+\beta)(R_f + R_e)$
 则可以用 $V_{CC} \frac{R_{b1}}{R_{b1} + R_{b2}}$ 来估计基极静态电压

$$\frac{2V - 0.7V}{1300\Omega} = 1mA$$

$$i_B = \frac{1mA}{1+\beta} = 10\mu A$$

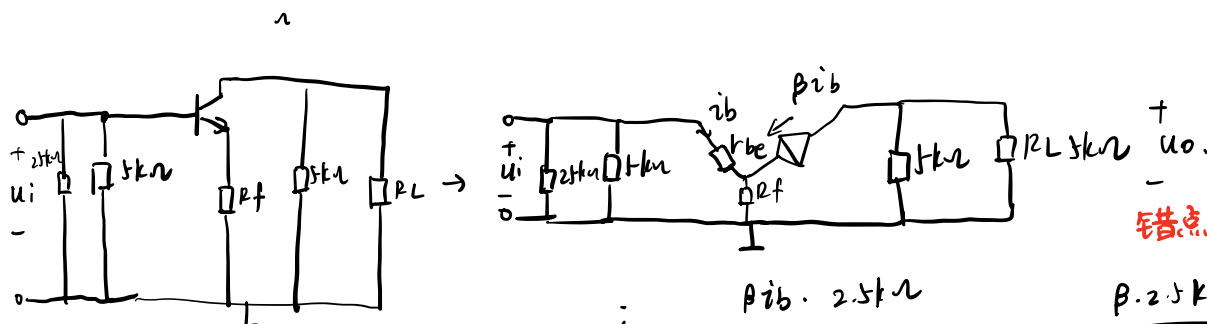
$$i_c \approx 1mA$$

$$U_{CE} = 5.7V$$

错点二： U_{CE} 不一定等于 U_o
 就像这个题目

$$U_{CE} + I_e \cdot (R_f + R_e) = U_o$$

交流通路



$$r_{be} = r_{bb'} + (1+\beta) \cdot \frac{U_T}{I_{EQ}}$$

$$= 100 + 101 \times \frac{26mV}{1mA} = 2726\Omega$$

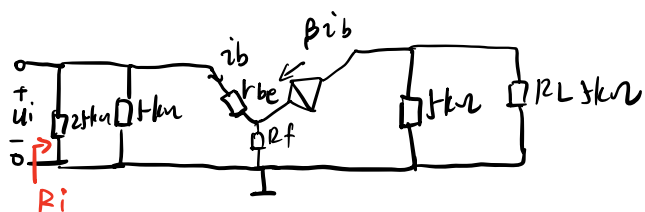
$$\dot{A}_u = - \frac{\beta i_b \cdot 25k\Omega}{i_b \cdot r_{be} + (1+\beta) i_b \cdot R_f} = - \frac{\beta \cdot 25k\Omega}{r_{be} + (1+\beta) R_f}$$

$$= - \frac{100 \times 25k\Omega}{2726\Omega + 101 \cdot 300\Omega}$$

$$= - \frac{250}{2.726 + 30.3} = -7.57$$

错点三： A_u 总是忘记要加负号

R_i



$$\frac{25k\Omega \times 5k\Omega}{25k\Omega + 5k\Omega} = \frac{125}{30} k\Omega = \frac{25}{6} k\Omega$$

设输入电流为 $1mA$

$$(1 - i_b) \times \frac{25}{6} k\Omega = i_b \cdot r_{be} + (1 + \beta) i_b \cdot R_f$$

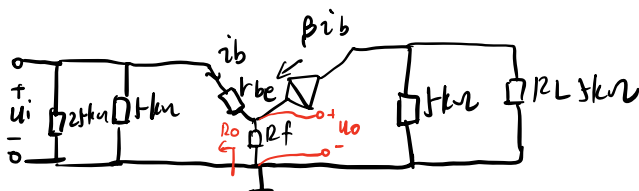
$$(1 - i_b) \times \frac{25}{6} k\Omega = 2.73 k\Omega \times i_b + 101 i_b \cdot 0.3 k\Omega$$

$$\frac{25}{6} - \frac{25}{6} i_b = 2.73 i_b + 30.3 i_b$$

$$i_b = \frac{\frac{25}{6}}{\frac{25}{6} + 2.73 + 30.3} = 0.112 mA$$

$$R_i = (1 - 0.112) \times \frac{25}{6} = 3.7 k\Omega$$

R_o



$$u_i = 0$$

$$i_b = 0$$

$$\beta i_b = 0 \text{ 断路}$$

$$R_o = 5k\Omega$$

(2)

$$\beta = 200$$

$$(1 + \beta)(R_f + R_e) \gg R_{b1} // R_{b2}$$

$$\therefore V_{\text{基极}} \approx 2V$$

$$I_{EQ} \text{ 仍为 } 1mA$$

$$I_{BQ} \approx \frac{1}{200} mA = 5\mu A$$

$$U_{CEQ} = 5.7V$$

(3) 这样交流通路 $\Rightarrow R_f \uparrow$

多一个 R_e

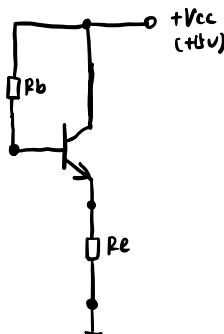
$|A_u| \downarrow$ A_u 放大倍数减小

$R_i \uparrow$

R_o 不变

2.12

(1) 直流通路



注意: 如果射极接了负载

$$i_b = \frac{V_{cc}}{R_b}$$

$$15V - R_b \cdot i_b - 0.7V - 81 \cdot i_b \cdot R_e = 0$$

$$14.3 = 200k\Omega \cdot i_b + 243k\Omega \cdot i_b$$

$$= 443k\Omega \cdot i_b$$

$$i_b = \frac{14.3}{443} = 0.03228mA$$

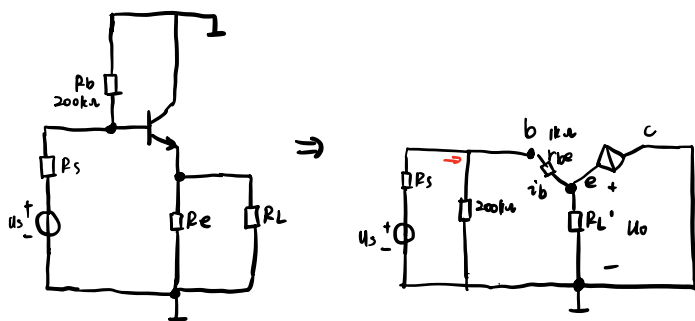
$$I_{BQ} = 0.03228mA$$

$$I_{CQ} = 2.58239mA$$

$$I_{EQ} = 2.61468mA$$

$$U_{CEQ} = 15V - 2.61468mA \times 3k\Omega = 7.15596V$$

(2) 交流通路



$$\textcircled{1} A_u = \frac{u_o}{u_i} = \frac{(\beta+1) \cdot R_L'}{(\beta+1) \cdot R_L' + r_{be}} = \frac{(\beta+1) \cdot R_L'}{(\beta+1) \cdot R_L' + r_{be}}$$

$R_L = \infty$ 时

$$A_u = \frac{81 \times 3}{81 \times 3 + 1} = 0.996$$

$R_L = 3k\Omega$ 时

$$A_u = 0.992$$

$\textcircled{2} R_i$: 当 $R_L = \infty$ 时

$$(1-i_b) \cdot 200k = i_b \cdot 1k + 81 \cdot i_b \cdot 3k$$

$$200 - 200i_b = 244i_b$$

$$200 = 444i_b$$

$$i_b = 0.45A$$

$$R_i = 0.55 \times 200k = 110k\Omega$$

当 $R_L = 3k\Omega$ 时

$$(1-i_b) \cdot 200k = i_b \cdot 1k + 81 \cdot i_b \cdot 1.5k$$

$$200 = (200 + 1 + 81 \times 1.5) i_b$$

$$i_b = 0.62A$$

$$R_i = (1 - 0.62) \times 200k = 76k\Omega$$

③ R_o 37Ω

$3k\Omega$ (不管 R_L 是 $3k\Omega$ 还是 ∞)

错点: 忽略右边恒流源

错点: 要搞清楚到底要不要把流控电流源放入讨论范围

$$\frac{3k}{81} = 37\Omega$$

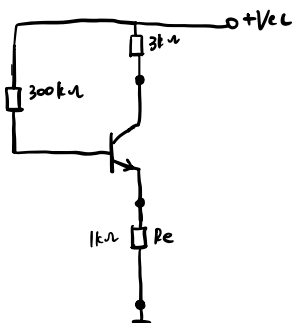
一般地, 若以集电极作为输出讨论 R_o 时, 可以直接不管流控电流源, 因为集电极加电压并不会影响基极电流 $U_i=0$ 时 i_b 仍为 0

但若以射极作为输出, 需考虑流控电流源, 因为在射极施加电压, 就算 $U_i=0$, 也会源发出 i_b , 因此计算 R_o 需要考虑流控电流源

2.13

(1) $\beta = 60$ $r_{bb'} = 100\Omega$

直流通路:



Q点:

$$12V = i_b \cdot 300k\Omega + 0.7V + 1k\Omega \cdot 61i_b$$

$$11.3V = 361k\Omega \cdot i_b$$

$$i_b = 0.0313mA = I_{BQ}$$

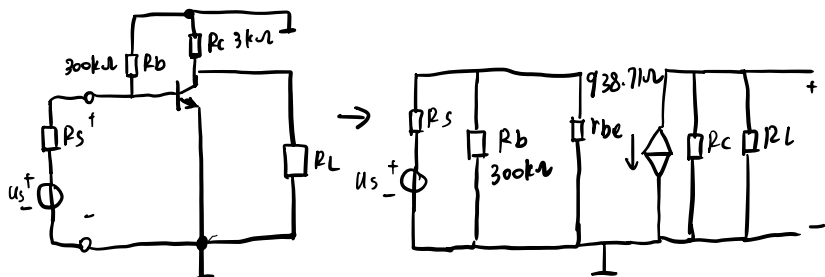
$$i_c = \frac{678}{361}mA = 1.878mA = I_{CQ}$$

$$U_{CEQ} = V_{CC} - 3k\Omega \times 1.878mA - 1k\Omega \cdot 61 \times 0.0313mA$$

$$= 4.4567V$$

$$r_{be} = r_{bb'} + (1+\beta) \frac{U_T}{I_{EQ}} \approx 61 \times \frac{26mV}{0.0313mA \times 61} = 938.71\Omega$$

交流通路:



$$A_u = - \frac{1.5k\Omega \times 60 \times i_b}{938.71 \cdot i_b} = -95.876$$

$$R_i \approx 938.71\Omega$$

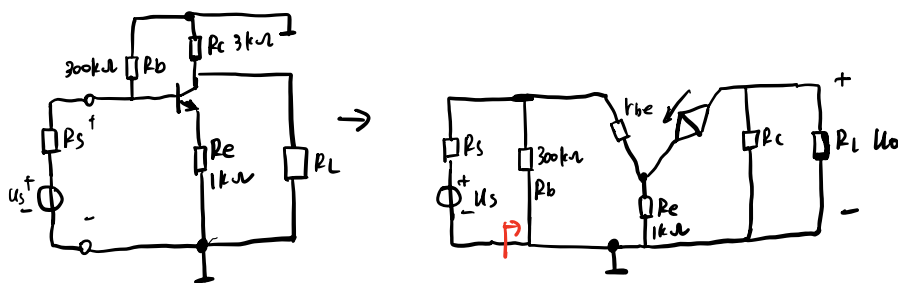
$$R_o = 1.5k\Omega \quad (R_o \text{ 是不看 } R_L \text{ 的 } R_o \text{ 与负载无关})$$

(2) ① $U_i = 10\text{mV} \times \frac{938.71\Omega}{2\text{k}\Omega + 938.71\Omega} = 3.19\text{mV}$

$U_o = -95.876 \times 3.19\text{mV} = -306.256\text{mV}$ 讨论的是有效值部分

② C_3 开路各时

错点: 是 C_3 开路而不是 R_L 输出端开路



$r_{be} = 938.71\Omega$

$A_u = \frac{U_o}{U_i} = - \frac{\beta i_b \cdot (R_c // R_L)}{i_b \cdot r_{be} + (\beta + 1) R_e}$

$= - \frac{60 \cdot 1.5\text{k}\Omega}{0.9387\text{k}\Omega + 61 \cdot 1\text{k}\Omega} = -1.45 \approx -1.5$

$300 \cdot (1 - i_b) = 0.9387\text{k}\Omega \cdot i_b + 1\text{k}\Omega \cdot 61 i_b$

$300 = (300 + 0.9387 + 61) i_b$

$i_b = 0.8288696\text{A}$

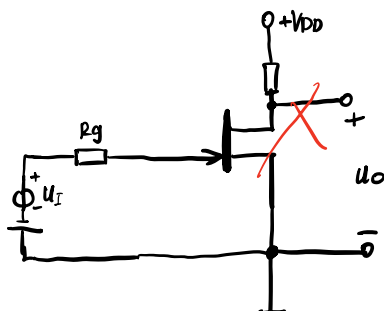
$R_i = 300\text{k}\Omega (1 - 0.8288696) = 51.33912\text{k}\Omega$

$U_i = 10\text{mV} \times \frac{51.33912}{2 + 51.33912} = 9.625\text{mV} \approx 9.6\text{mV}$

$U_o = 9.6\text{mV} \times (-1.5) = -14.4\text{mV}$

2.14

(a) 没有给偏置电压, 导致 U_{GS} 会大于 0



在栅极加上 R_s

(b) N沟道耗尽型场效应管 交流通路中
交流电压源短路, 交流电流源断路

需要在漏极加电阻 R_d , 从而将电流变化 $\rightarrow$ 电压变化

在输入端加入电容

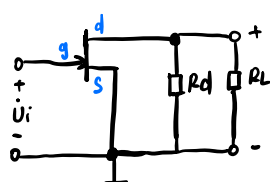
(c) V_{DD} 改为 $-V_{DD}$

且 R_g 支路加上 $-V_{GS}$

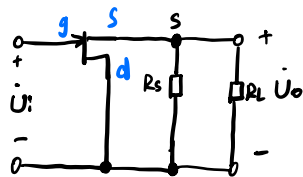
2.15

场效应管放大电路分析

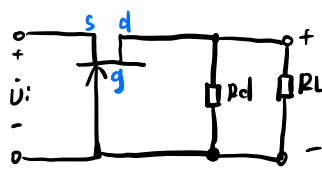
知识点一: 场效应管放大电路三种接法



(a) 共源放大电路



(b) 共漏放大电路

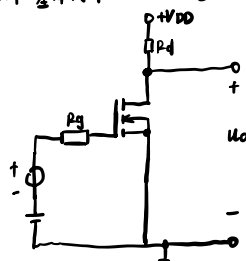


(c) 共栅极放大电路

知识点二: 场效应管静态工作点的设置

原则: 使场效应管工作在恒流区

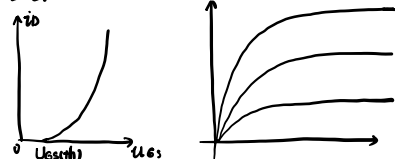
例: 基本共源放大电路



1. 判断管子:

N沟道增强型场效应管

2. 画出特征曲线



3. 找工作在恒流区条件

$U_{GS} > U_{GS(th)}$

$U_{DS} > U_{GS} - U_{GS(th)}$

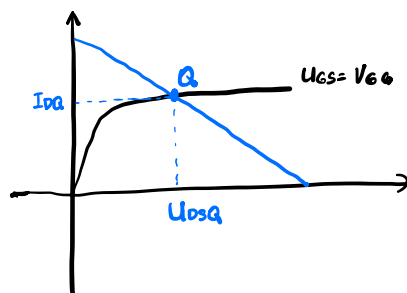
4. 具体各条件的实现方式

输入回路加 V_{GS}

漏极电源

★ R_d 作用: 将漏极电流 i_D 的变化, 转换为 u_{DS} 的变化, 进而实现电压放大

5. 图解法



负载线:

$U_{DS} = V_{DD} - I_D \cdot R_D$

$I_D = \frac{V_{DD}}{R_D} - \frac{1}{R_D} U_{DS}$

6. 利用场效应管电流方程求 I_{DQ}

$$i_D = I_{D0} \left(\frac{U_{GS}}{U_{GS(th)}} - 1 \right)^2 \quad \text{针对不同管子, } I_{D0} \text{ 定义不一样}$$

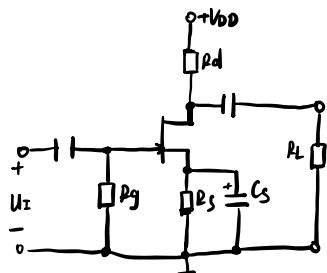
在这里, 针对 N 沟道型增强型 I_{D0} 为 $U_{GS} = 2 U_{GS(th)}$ 的时候对应的电流

在这里:

$$I_{DQ} = I_{D0} \cdot \left(\frac{V_{GS}}{U_{GS(th)}} - 1 \right)^2$$

$$U_{DSQ} = V_{DD} - I_{DQ} \cdot R_d$$

知识点三: 自给偏压电路



问: 这个电路只有在 $U_{GS} < 0$ 时才会正常工作, 那么这个电路的 U_{GS} 为什么会小于 0?

解释: 栅极绝缘

$$I_g = 0$$

$$U_{GQ} = 0$$

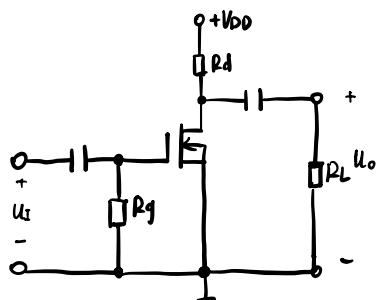
而 I_{DQ} 流过源极必然产生电压

$$U_{SQ} = I_{DQ} \cdot R_s$$

$$U_{GS} = -I_{DQ} \cdot R_s < 0$$

∴ 实际上电路是靠源极电阻上的电压为栅源两极提供一个负偏压, 故称为自给偏压

自给偏压电路的特例:

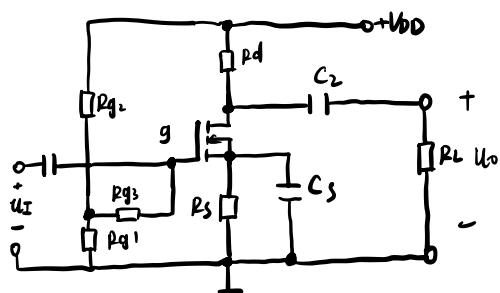


这里 $U_{GSQ} = 0$. 图中采用耗尽型 N 沟道 MOS 管

其栅源电压在小于 0, 等于 0, 大于 0 的一定范围内均可正常工作

∴ 可行

知识点四: 分压式偏置电路



在这里

静态时

栅极电流为 0, 所以电阻 R_{g3} 上电流为 0

栅极电位和源极电位分别为:

$$U_{GQ} = V_{DD} \times \frac{R_{g1}}{R_{g1} + R_{g2}}$$

$$U_{SQ} = I_{DQ} \cdot R_s$$

$$\therefore U_{GSQ} = \frac{R_{g1}}{R_{g1} + R_{g2}} V_{DD} - I_{DQ} \cdot R_s$$

$$\text{与 } I_{DQ} = I_{D0} \left(\frac{U_{GSQ}}{U_{GS(th)}} - 1 \right)^2 \text{ 联立即可求解}$$

知识点五：场效应管的低频小信号等效模型

原理： $i_D = f(u_{GS}, u_{DS})$

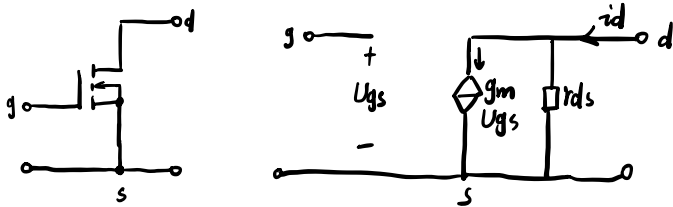
$$di_D = \left. \frac{\partial i_D}{\partial u_{GS}} \right|_{u_{DS}} \cdot du_{GS} + \left. \frac{\partial i_D}{\partial u_{DS}} \right|_{u_{GS}} \cdot du_{DS}$$

$$\frac{\partial i_D}{\partial u_{GS}} = g_m \Rightarrow \text{低频跨导, 表示场效应管 } i_D \text{ 受 } u_{GS} \text{ 控制情况}$$

$$\frac{\partial i_D}{\partial u_{DS}} = \frac{1}{r_{DS}}$$

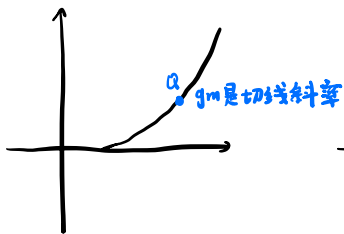
$$\Delta I_D = g_m \cdot \Delta U_{GS} + \frac{1}{r_{DS}} \cdot \Delta U_{DS}$$

等效模型原理图

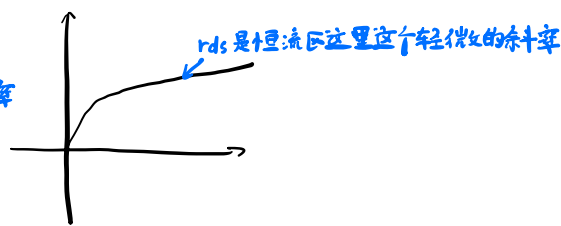


求取 g_m 与 r_{DS}

转移特性：



输出特性



对增强型的 g_m

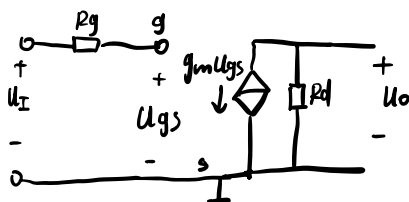
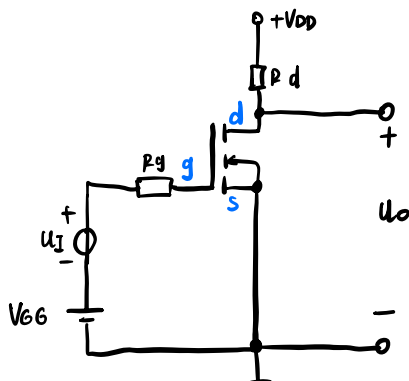
$$i_D = I_{D0} \left(\frac{u_{GS}}{U_{GS(th)}} - 1 \right)^2 \quad \frac{u_{GS}}{U_{GS(th)}} - 1 = \sqrt{\frac{i_D}{I_{D0}}}$$

$$g_m = \left. \frac{\partial i_D}{\partial u_{GS}} \right|_{u_{DS}} = \frac{2I_{D0}}{U_{GS(th)}} \left(\frac{u_{GS}}{U_{GS(th)}} - 1 \right) \Big|_{u_{DS}} = \frac{2}{U_{GS(th)}} \cdot I_{D0} \cdot \sqrt{\frac{i_D}{I_{D0}}}$$

$$\approx \frac{2}{U_{GS(th)}} \times \sqrt{I_{D0} \cdot I_{DQ}} \quad (\text{用 } I_{DQ} \text{ 来估计 } i_D)$$

知识点六：电路分析

A 共源放大电路的动态分析

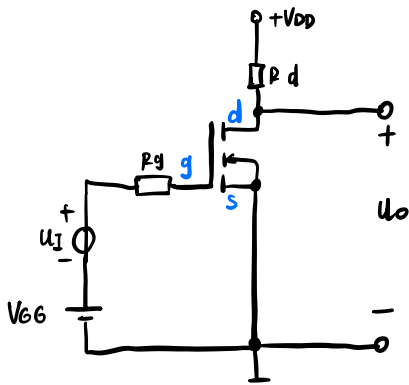


$$\dot{A}_u = \frac{u_o}{u_i} = - \frac{i_d \cdot R_d}{u_{gs}} = - \frac{g_m \cdot u_{gs} \cdot R_d}{u_{gs}} = -g_m \cdot R_d$$

$$R_i = \infty$$

$$R_o = R_d$$

例题: $V_{GG} = 6V$, $V_{DD} = 12V$, $R_d = 3k\Omega$; 场效应管开启电压 $U_{GS(th)} = 4V$, $I_{DQ} = 10mA$ 。试估算电路的Q点, $\dot{A}_u$ 和 R_o 。



这是N沟道增强型场效应管

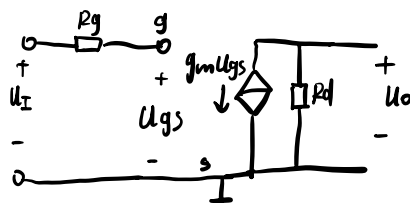
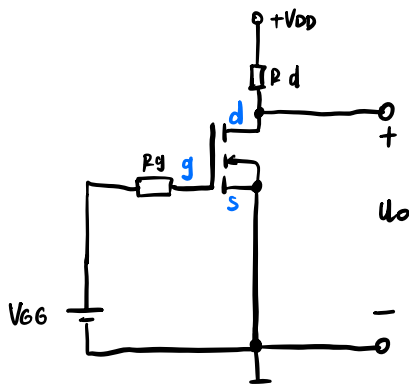
$$i_D = I_{DQ} \left(\frac{U_{GS}}{U_{GS(th)}} - 1 \right)^2$$

$$U_{GS} = V_{GG} = 6$$

$$I_{DQ} = 10mA \cdot \left(\frac{6}{4} - 1 \right)^2$$

$$= 2.5mA$$

$$U_{DSQ} = 12V - 3k\Omega \cdot 2.5mA = 4.5V$$



$$g_m = \frac{\partial i_D}{\partial u_{gs}}$$

$$g_m = \frac{2I_{DQ}}{U_{GS(th)}} \left(\frac{U_{GS}}{U_{GS(th)}} - 1 \right)$$

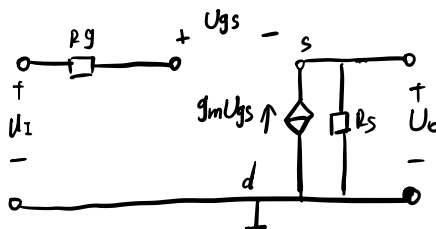
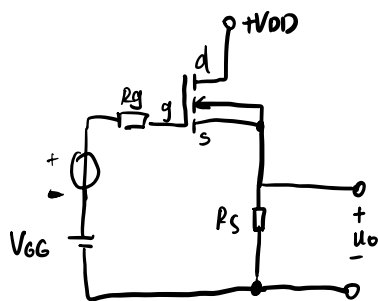
$$= \frac{20mA}{4V} \times \left(\frac{6}{4} - 1 \right)$$

$$= 5mA/V \times 0.5 = 2.5mA/V = 2.5mS$$

$$\dot{A}_u = \frac{u_o}{u_i} = - \frac{g_m \cdot u_{gs} \cdot R_d}{u_{gs}} = -3k\Omega \cdot 2.5mS = -7.5$$

$$R_o = 3k\Omega \quad (\text{把 } U_i \text{ 设置为 } 0, \text{ 然后 } u_{gs} = 0, g_m u_{gs} = 0, \text{ 整个 = 断路})$$

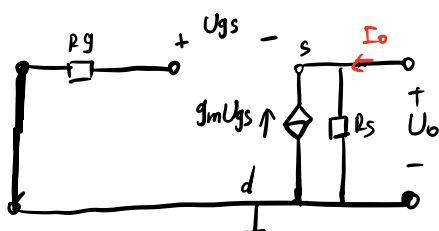
B, 基本共漏放大电路的动态分析



$$A_u = \frac{g_m \cdot U_{gs} \cdot R_S}{U_{gs} + g_m \cdot U_{gs} \cdot R_S} = \frac{g_m \cdot R_S}{1 + g_m \cdot R_S}$$

$$R_i = \infty$$

$$R_o = U_i = 0$$



$$I_D = \frac{U_o}{R_S} + g_m U_o$$

$$\frac{I_o}{U_o} = \frac{1}{R_S} + \frac{1}{\frac{1}{g_m}}$$

$$= \frac{R_S + \frac{1}{g_m}}{R_S \cdot \frac{1}{g_m}}$$

$$\frac{U_o}{I_o} = R_S \parallel \frac{1}{g_m}$$

例题: $U_{GS(th)} = 3V$ $I_{DQ} = 8mA$ $R_S = 3k\Omega$ $I_{DQ} = 2.5mA$

计算电路的 A_u , R_i , R_o

$$g_m = \frac{2}{U_{GS(th)}} \sqrt{I_{DQ} \cdot I_{DQ}} \approx 2.98mS$$

$$A_u = \frac{U_o}{U_i} = \frac{g_m \cdot R_S}{1 + g_m \cdot R_S} = \frac{2.98mS \times 3k\Omega}{1 + 2.98mS \times 3k\Omega} = 0.8994$$

$$R_i = \infty$$

$$R_o = 3k\Omega \parallel \frac{1}{2.98mS} \approx 302\Omega$$

2.15

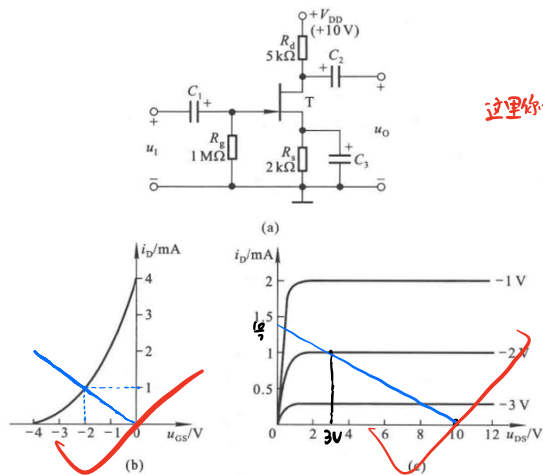
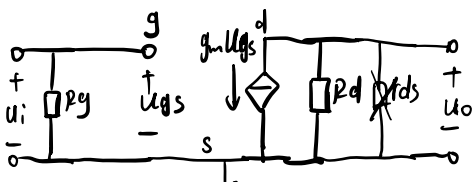
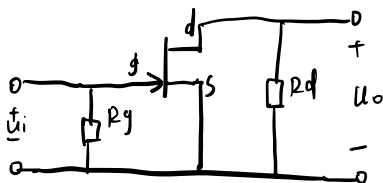
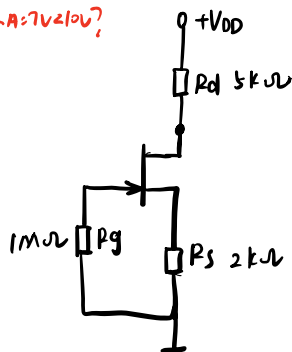


图 P2.15

交流通路



直流通路



这里你看 \$7k \cdot 1\text{mA} = 7\text{V} < 10\text{V}\$?
为啥

$$u_{DS} = 10 - 5\text{ k}\Omega \cdot i_D$$

$$u_{GS} = -2\text{ k}\Omega \cdot i_D$$

$$i_D = 2\text{ mA}$$

$$u_{GS} = -4\text{ V}$$

$$I_{DQ} = 1\text{ mA}$$

$$U_{GSQ} = -2\text{ V}$$

$$U_{DSQ} = V_{DD} - R_D \cdot I_{DQ} - R_S \cdot I_{DQ} = 3\text{ V}$$

问: 为什么会经过
1.5 mA 而不是
1 mA 呢?

$$A_u = \frac{u_o}{u_i} = -\frac{g_m u_{gs} \cdot R_d}{u_{gs}} = -g_m \cdot R_d$$

$$i = 4\text{ mA} \cdot \left(1 - \frac{u_{gs}}{u_{gs(\text{off})}}\right)^2$$

$$\frac{di}{du_{gs}} = 8\text{ mA} \cdot \left(-\frac{1}{u_{gs(\text{off})}}\right) \cdot 2 \cdot \left(1 - \frac{u_{gs}}{u_{gs(\text{off})}}\right)$$

$$= 8\text{ mA} \cdot \left(-\frac{1}{4}\right) \cdot 2 \cdot \left(1 - \frac{1}{2}\right)$$

$$= -1\text{ mA} = -g_m$$

$$A_u = -g_m \cdot R_d = -1\text{ mA} \cdot 5\text{ k}\Omega = -5$$

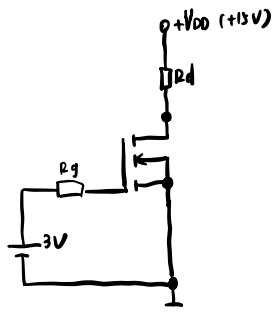
$$R_i = R_g = 1\text{ M}\Omega$$

$$R_o = R_d = 5\text{ k}\Omega$$

错点: 处理2次求导一定要小心,
2倍只乘1次

2.16

直流通路



$$U_{GS} = 3V$$

$$i_D = i_{D0} \left(\frac{U_{GS}}{U_{GS(th)}} - 1 \right)^2$$

$$i_D = 4mA \cdot \left(\frac{3V}{2V} - 1 \right)^2$$

$$U_{GS} = 3V$$

$$i_D = 4mA \cdot \left(\frac{3}{2} - 1 \right)^2 = 1mA$$

$$Q点: I_{DQ} = 1mA$$

$$U_{GSQ} = 3V$$

$$U_{DSQ} = 5V$$

$$U_{GSQ}$$

$$i_D = 4 \left(\frac{U_{GS}}{U_{GS(th)}} - 1 \right)^2$$

$$g_m = 8 \cdot \frac{1}{U_{GS(th)}} \left(\frac{U_{GS}}{U_{GS(th)}} - 1 \right)$$

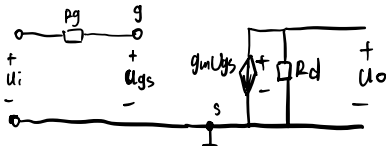
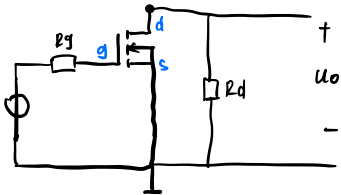
$$g_m = 8 \cdot \frac{1}{2} \cdot \left(\frac{3}{2} - 1 \right) = 2mS$$

错点:

场效应管里面没有c.e两板

只有g, d, s 这几种

交流通路:

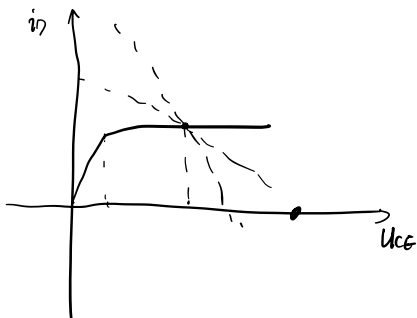


$$\dot{A}_u = \frac{U_o}{U_i} = -g_m \cdot R_d = -2mS \cdot 10k\Omega = -20$$

$$R_i = \infty$$

$$R_o = R_d = 10k\Omega$$

2.17



$$V_{DD} - I_D \cdot R_d = U_{CE}$$

$$I_D = \frac{V_{DD}}{R_d} - \frac{U_{CE}}{R_d}$$

$$\frac{1}{R_d} \downarrow \quad R_d \uparrow$$

(1)

① 底部

增大 R_1/R_2 减小 R

增大 V_{DD} / 增大 R_2 / 减小 R_1

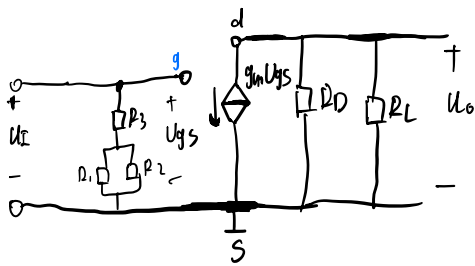
② 顶部失真 $\Rightarrow$ 减小 R_1, R_2 , 增大 R_2

问: 截止到底是

还是



(2) 交流通路



提高 R_{cl}

~~减小 R_S~~

g_m : 让Q点右移 $\Leftarrow U_{gs} \uparrow$ 减小 R_1, R_2 , 增大 R_2

2.18

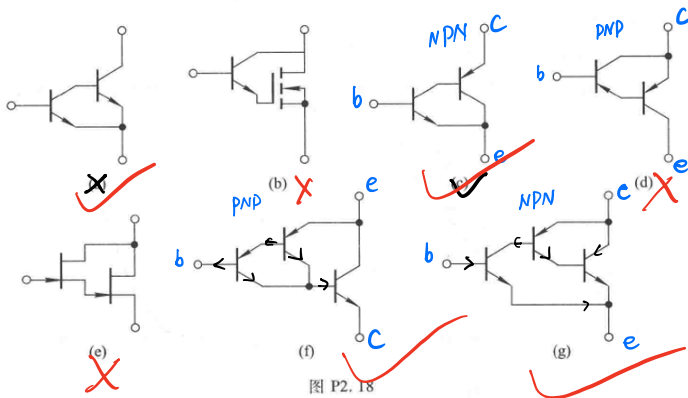


图 P2.18

因为无集电极
电流通路 ??